

Industrial Policies Cascade in Production Networks

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Motivation

- Industrial policy is back at the center of policymaking: CHIPS Act, IRA, EU green-industrial agenda, China's sectoral programs
 - Which sectors generate the largest aggregate return to targeted intervention?
- Liu (2019): *distortion centrality* ranks sectors by the intensive-margin gain from narrowing wedges $\Rightarrow$ points *upstream*, where wedges compound
- Yet much observed policy targets large *downstream* markets: automobiles, transportation equipment, housing, construction
- Missing margin: subsidies also change **market structure** — sales, the mass of firms, the set of varieties, and (under variable markups) competitive pressure
 - Entry cascades in production networks amplify shocks (Baqaei, 2018; Taschereau-Dumouchel, 2026; Elliott et al., 2022)

This Paper

- A production-network economy with imperfect competition and endogenous firm entry and exit
 - General framework of Baqaee and Farhi (2020), Bigio and La'O (2020) + free entry with Kimball demand, following Baqaee (2018)
 - Entry affects output through *returns to variety* (η_k) and a *pro-competitive* markup response (ξ_k)
- The first-order output response to a sectoral subsidy decomposes into two sufficient statistics:
 - **Distortion centrality** (intensive margin): gain from alleviating distortions, as in Liu (2019)
 - **Entry centrality** (extensive margin, new): output value of the cascading firm entry a subsidy triggers across the network
- We characterize the equilibrium entry cascade in closed form as a *network multiplier*

Preview of Results

- **Theory:** $d \log Y / d \log \tau_k$ = distortion centrality + entry centrality; the entry response solves a network fixed point with closed-form solution
- **U.S. calibration (2019, 66 sectors):** entry centrality *re-ranks* sectors rather than refining the ranking
 - $\text{corr}(\text{distortion}, \text{entry centrality}) = -0.40$; distortion centrality loads upstream (corr. with upstreamness +0.89), entry centrality downstream (-0.76)
 - High entry centrality: housing, financial vehicles, food and beverage, motor vehicles
- **Magnitude:** the extensive margin accounts for between one quarter and one half of the aggregate first-order output response (45% at the benchmark) — a share pinned down by the input–output data even though levels are not
- **China (2007, 135 sectors):** the two centralities are negatively correlated (-0.79); observed interventions load on both — adding entry centrality raises R^2 for effective interest rates from 0.343 to 0.556

Firms: Technologies

- Firm ν in sector k uses labor and intermediates with CRS technology

$$y_k(\nu) = z_k F(l_k(\nu), \{x_{kh}(\nu)\}_{h=1}^N)$$

- Conditional on entry, firms minimize marginal cost,

$$mc_k = \min_{\{x_{kh}(\nu)\}, l_k(\nu)} W l_k(\nu) + \sum_{h=1}^N P_h x_{kh}(\nu),$$

subject to a unit production constraint

- Sectoral output Y_k aggregates varieties through a homothetic **Kimball** aggregator

$$\int_0^{M_k} \Upsilon_k(q_k(\nu)) d\nu = 1, \quad q_k(\nu) \equiv \frac{y_k(\nu)}{Y_k},$$

with M_k the mass of firms; Dixit–Stiglitz is the special case $\Upsilon_k(q) = q^{(\varepsilon_k-1)/\varepsilon_k}$

Firms: Entry Moves Prices and Wedges

- Pricing: desired markup times subsidy-adjusted marginal cost,

$$p_k(v) = \frac{\varepsilon_k}{\varepsilon_k - 1} \tau_k^{-1} mc_k = \mu_k mc_k, \quad \varepsilon_k = -\frac{\Upsilon'_k(q_k)}{q_k \Upsilon''_k(q_k)}$$

μ_k is the sectoral **wedge**; Kimball demand lets ε_k vary with relative scale q_k

- Entry changes each producer's relative scale q_k ; two elasticities summarize the response:

- Returns to variety:** $\eta_k \equiv -\frac{d \log P_k}{d \log M_k}$ (price index falls with entry)
- Pro-competitive elasticity:** $\xi_k \equiv -\frac{d \log \mu_k}{d \log M_k} \geq 0$ (entry compresses the wedge; sign from Marshall's Second Law of Demand)

- Dixit–Stiglitz: $\eta_k = 1/(\varepsilon_k - 1)$ constant and $\xi_k = 0$

Firms: Overhead and Free Entry

- Firms pay a fixed **overhead** $f_k = \phi_k \sum_{j=1}^N P_j C_j$ to operate (ϕ_k : structural entry barrier); profits

$$\pi_k(v) = p_k(v)y_k(v) - \tau_k^{-1}mc_k y_k(v) - f_k$$

- With symmetric CRS firms, operating profits are $\pi_k(v) = \frac{1}{M_k} \frac{1}{\varepsilon_k} P_k Y_k - f_k$, so free entry ($\pi_k = 0$) pins the mass of firms to the *sales share* λ_k :

$$M_k = \frac{1}{\varepsilon_k} \frac{P_k Y_k}{f_k} = \frac{\lambda_k}{\varepsilon_k \phi_k}$$

Entry is an equilibrium object: it rises with market size and the operating margin

- Baseline: the overhead is **rebated** lump-sum to final users — it disciplines entry but absorbs no resources (overhead paid in labor: Online Appendix)

Government and Household

- **Policy instrument:** sector-specific subsidies $\{\tau_k\}_{k=1}^N$ scale marginal costs from mc_k to $\tau_k^{-1}mc_k$ ($\tau_k > 1$: subsidy; $\tau_k < 1$: tax), financed by a lump-sum tax

$$T = \sum_{k=1}^N (1 - \tau_k^{-1}) mc_k \int_0^{M_k} y_k(v) dv$$

- **Final demand:** real output $Y = \max \mathcal{Y}(\{C_k\}_{k=1}^N)$, a constant-returns aggregator, subject to

$$\sum_{k=1}^N P_k C_k = W\bar{L} + \sum_{k=1}^N \int_0^{M_k} [\pi_k(v) + f_k] dv - T$$

with inelastic labor endowment $\bar{L} = 1$

- **Equilibrium:** firms optimize; final uses maximize Y ; government budget holds; goods and labor markets clear; free entry implies zero profits

Notation: Two Input–Output Matrices

Labor is industry $L = N + 1$ (endowment producer)

- Cost-based (forward) and revenue-based (backward) input–output matrices

$$\tilde{\Omega}_{kh} = \frac{\partial \log mc_k}{\partial \log P_h} = \frac{P_h X_{kh}}{\mu_k^{-1} P_k Y_k}, \quad \Omega = \text{diag}(\mu)^{-1} \tilde{\Omega}$$

They differ exactly where wedges create a gap between revenues and costs

- Leontief inverses and Domar weights

$$\tilde{\Psi} = (I - \tilde{\Omega})^{-1}, \quad \Psi = (I - \Omega)^{-1}, \quad \tilde{\lambda}' = b' \tilde{\Psi}, \quad \lambda' = b' \Psi$$

- $\Lambda_L = W\bar{L} / \sum_{j=1}^N P_j C_j$: **labor income share** = Domar weight of labor
 - Frictionless economy: $\Lambda_L = 1 \Rightarrow \Lambda_L$ measures the *aggregate inverse wedge*
 - Ψ_{iL} : inverse wedge accumulated along sector i 's supply chain (= 1 without wedges)

The Substitution Matrix

- Nested-CES production and consumption with cross-sector elasticities $\sigma = \{\sigma_j\}_{j=0}^N$ (σ_0 : consumption; σ_k : production)
- Input–output covariance operator (Baqaei and Farhi, 2020): $(N + 1) \times (N + 1)$ matrix

$$[\Phi(\sigma)]_{l,k} = \sum_{j=0}^N \sigma_j \lambda_j \mu_j^{-1} \text{Cov}_{\tilde{\Omega}(j,:)} \left(\frac{\Psi_{(l)}}{\lambda_l}, \frac{\tilde{\Psi}_{(k)}}{\tilde{\lambda}_k} \right)$$

- Φ is *additively linear* in the elasticities: $\Phi(u) + \Phi(v) = \Phi(u + v)$
- Cobb–Douglas benchmark ($\sigma = \mathbf{1}$):

$$[\Phi(\mathbf{1})]_{L,k} = \frac{\lambda_k}{\tilde{\lambda}_k} \frac{\Psi_{kL}}{\Lambda_L} - 1$$

Aggregation Theorem

Theorem

In response to shocks to productivities and subsidies, the first-order change in real output is

$$d \log Y = \underbrace{\sum_{k=1}^N \tilde{\lambda}_k d \log z_k}_{\Delta \text{ technology}} + \underbrace{\sum_{k=1}^N \tilde{\lambda}_k \eta_k d \log M_k}_{\Delta \text{ variety}} - \underbrace{\sum_{k=1}^N \tilde{\lambda}_k d \log \mu_k - d \log \Lambda_L}_{\Delta \text{ allocative efficiency}}$$

Following changes in sectoral subsidies,

$$d \log Y = \underbrace{\sum_{k=1}^N \left(\tilde{\lambda}_k + \frac{d \log \Lambda_L}{d \log \mu_k} \right) d \log \tau_k}_{\text{intensive margin}} + \underbrace{\sum_{k=1}^N \left[\eta_k \tilde{\lambda}_k + \xi_k \left(\tilde{\lambda}_k + \frac{d \log \Lambda_L}{d \log \mu_k} \right) - \frac{d \log \Lambda_L}{d \log M_k} \right] d \log M_k}_{\text{extensive margin}}$$

- Hulten + a direct variety effect ($d \log P_k = -\eta_k d \log M_k$) + reallocation summarized by the labor income share

The Labor Income Share Responds to Wedges and Entry

Lemma

$$\frac{d \log \Lambda_L}{d \log \mu_k} = -\tilde{\lambda}_k - \tilde{\lambda}_k [\Phi(\sigma)]_{L,k} \quad \text{and} \quad \frac{d \log \Lambda_L}{d \log M_k} = \eta_k \tilde{\lambda}_k [\Phi(\sigma - \mathbf{1})]_{L,k}$$

- The labor row of the substitution matrix is the *network kernel* of reallocation
- Without wedges, $[\Phi(\sigma)]_{L,k} = 0$; Leontief ($\sigma = \mathbf{0}$): $[\Phi(\mathbf{0})]_{L,k} = 0$
- Horizontal economy: $[\Phi(\mathbf{1})]_{L,k} = \bar{\mu}/\mu_k - 1$ with $\bar{\mu} = [\mathbb{E}_b(\mu^{-1})]^{-1} = 1/\Lambda_L$ the aggregate wedge $\Rightarrow [\Phi(\mathbf{1})]_{L,k}$ measures *relative distortion* (negative = above-average wedge)

Corollary (Sectoral entry and aggregate distortion)

If $\sigma_j = \sigma$ for all j , then $\frac{d \log \Lambda_L}{d \log M_k} = (\sigma - 1) \eta_k \tilde{\lambda}_k [\Phi(\mathbf{1})]_{L,k}$: entry lowers aggregate market power iff $(\sigma - 1)[\Phi(\mathbf{1})]_{L,k} > 0$ (cf. Edmond et al., 2023)

Two Sufficient Statistics

Proposition

The output response to a sectoral subsidy decomposes into distortion centrality and entry centrality,

$$\frac{d \log Y}{d \log \tau_k} = \underbrace{\frac{\partial \log Y}{\partial \log \tau_k}}_{\text{distortion centrality}} + \underbrace{\sum_{l=1}^N \frac{\partial \log Y}{\partial \log M_l} \frac{d \log M_l}{d \log \tau_k}}_{\text{entry centrality}},$$

where

$$\frac{\partial \log Y}{\partial \log \tau_k} = -\tilde{\lambda}_k[\Phi(\sigma)]_{L,k}, \quad \frac{\partial \log Y}{\partial \log M_k} = \eta_k \lambda_k \frac{\Psi_{kL}}{\Lambda_L} - (\eta_k + \xi_k) \tilde{\lambda}_k[\Phi(\sigma)]_{L,k}$$

- Distortion centrality: output gain from narrowing sector k 's wedge, holding entry fixed (as in Liu, 2019, here with substitution elasticities as in Baqaee and Farhi, 2020)
- Entry centrality: aggregates the variety value and the pro-competitive wedge compression of the induced entry; $\eta_l = \xi_l = 0 \Rightarrow$ entry carries no first-order output effect

The Entry Cascade: Ingredients

- **Wedge**: responds to the subsidy and, through the pro-competitive channel, to entry

$$d \log \mu_k = -d \log \tau_k - \xi_k d \log M_k$$

- **Free entry**: the mass of firms tracks the sales share

$$(1 + (\varepsilon_k - 1) \xi_k) d \log M_k = d \log \lambda_k$$

- **Sales shares**: wedges propagate backward and forward; entry (given wedges) propagates only forward

$$d \log \lambda = - \underbrace{\left[\text{diag}(\lambda^{-1}) \Psi' \text{diag}(\lambda) - I \right] d \log \mu}_{\text{Backward propagation}} + \underbrace{\Phi(\sigma - \mathbf{1}) \text{diag}(\tilde{\lambda}) (\eta \circ d \log M - d \log \mu)}_{\text{Forward propagation}}$$

- Feedback loop: entry $\rightarrow$ prices $\rightarrow$ sales shares $\rightarrow$ entry $\Rightarrow$ subsidies set off **entry cascades**

The Entry Cascade Is a Network Multiplier

Proposition (Entry cascades from sectoral subsidies)

The response of the mass of entrants to subsidies is

$$\frac{d \log M}{d \log \tau} = \left[I - \Lambda_M + (D_\varepsilon - I - \Lambda_\mu) D_\xi \right]^{-1} \Lambda_\mu$$

where $\Lambda_\mu = -\frac{\partial \log \lambda}{\partial \log \mu} = \text{diag}(\lambda^{-1}) \Psi' \text{diag}(\lambda) - I + \Phi(\sigma - \mathbf{1}) \text{diag}(\tilde{\lambda})$ and

$\Lambda_M = \frac{\partial \log \lambda}{\partial \log M} = \Phi(\sigma - \mathbf{1}) \text{diag}(\tilde{\lambda} \circ \eta)$. Dixit–Stiglitz ($\xi_k = 0$) yields $(I - \Lambda_M)^{-1} \Lambda_\mu$.

- Λ_μ : impulse — the direct effect of subsidy-induced wedge changes on sales shares
- Λ_M : returns-to-variety feedback, compounding like input–output rounds
- D_ξ terms: pro-competitive feedback — own-entry *dampening* (thinner margins fund fewer firms) plus cross-sector propagation
- A *market-structure multiplier*: linkages transmit a competitive shock across markets

Transparent Benchmark: Cobb–Douglas with Dixit–Stiglitz Demand

- With $\sigma = 1$ (so $\Lambda_M = 0$) and $\xi_k = 0$, $\eta_k = 1/(\varepsilon_k - 1)$:

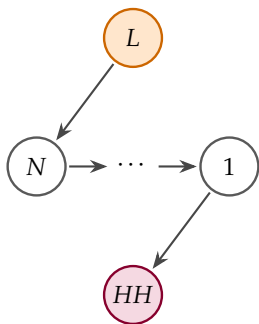
$$\frac{\partial \log Y}{\partial \log \tau_k} = \tilde{\lambda}_k - \lambda_k \frac{\Psi_{kL}}{\Lambda_L}, \quad \frac{\partial \log Y}{\partial \log M_l} = \frac{\tilde{\lambda}_l}{\varepsilon_l - 1}, \quad \frac{\partial \log M_l}{\partial \log \tau_k} = \frac{\lambda_k}{\lambda_l} (\Psi_{kl} - \delta_{kl})$$

so that

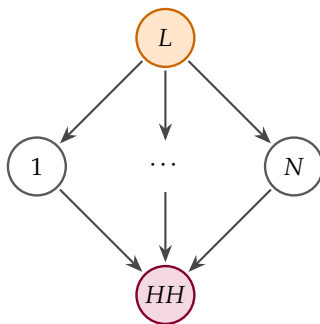
$$\frac{d \log Y}{d \log \tau_k} = \underbrace{\tilde{\lambda}_k - \lambda_k \frac{\Psi_{kL}}{\Lambda_L}}_{\text{distortion centrality}} + \underbrace{\lambda_k \sum_{l=1}^N \frac{\tilde{\lambda}_l}{\varepsilon_l - 1} \frac{\Psi_{kl} - \delta_{kl}}{\lambda_l}}_{\text{entry centrality}}$$

- A subsidy induces entry *backward* along the supply chain ($\Psi_{kl} > 0$ upstream)
- Uniform ε , no wedge gap: entry centrality per unit of sales $\propto \sum_{l=1}^N \Psi_{kl}$, the **backward supply-chain multiplier** — large for sectors whose expansion pulls strongly on their chains, not for upstream sectors themselves

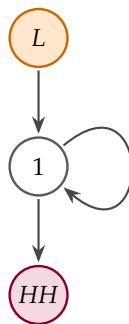
Three Example Economies



(A) Vertical



(B) Horizontal



(C) Roundabout

- Vertical: the entry cascade along a supply chain Horizontal: reallocation of entry through demand Roundabout: interaction of the two margins within one sector

Ex. Vertical Economy: the Cascade Is Directional

- Chain $N \rightarrow \dots \rightarrow 1 \rightarrow HH$; sector N uses only labor. Propositions 1–2 yield

$$\frac{\partial \log Y}{\partial \log \tau_k} = 0, \quad \frac{\partial \log Y}{\partial \log M_j} = \eta_j, \quad \frac{\partial \log M_j}{\partial \log \tau_k} = \begin{cases} \delta_j \prod_{k < h < j} (1 + \xi_h \delta_h), & j > k, \\ 0, & j \leq k, \end{cases}$$

where $\delta_k = \frac{1}{1 + (\varepsilon_k - 1)\xi_k}$

- One feasible allocation $\Rightarrow$ distortion centrality is *zero*; the whole response is entry
- The induced entry is **strictly upstream**: a subsidy to k passes its cost reduction downstream through prices (sales shares of $j \leq k$ unchanged), while *upstream* sales shares rise — free entry then draws firms in
- Variable markups *amplify* the cascade: ξ_h of the sectors between k and j compounds the response

Ex. Horizontal Economy: Reallocation through Demand

- Labor-only production; CES consumption with elasticity σ_0 . Distortion centrality:

$$\frac{\partial \log Y}{\partial \log \tau_k} = \sigma_0 b_k \mu_k^{-1} (\mu_k - \bar{\mu}), \quad \bar{\mu} = [\mathbb{E}_b(\mu^{-1})]^{-1}$$

positive iff sector k 's wedge exceeds the aggregate wedge $\bar{\mu}$

- Entry margin: a subsidy lowers P_k and reallocates expenditure — *away* from k under complementarity ($\sigma_0 < 1$), *toward* k under substitutability ($\sigma_0 > 1$)

Corollary (homogeneous wedges)

Distortion centrality = 0, and entry centrality = $(1 - \sigma_0) b_k \tilde{\chi}_k \left[\left(1 + (\varepsilon_k - 1) \xi_k \right) \frac{\mathbb{E}_b(\eta \tilde{\chi})}{\mathbb{E}_b(\omega)} - \eta_k \right]$

- Nonzero only when sectors differ in η_k or ξ_k : under Dixit–Stiglitz, $\eta_k = 1/(\varepsilon_k - 1)$ is uniform and $\xi_k = 0$, so entry centrality vanishes too — the extensive margin needs heterogeneity in returns to variety or the pro-competitive channel

Ex. Roundabout Economy: the Margins Interact

- One sector uses labor and its own output (CES, $\sigma_1 \in (0, 1)$); at $\tau_1 = 1$, $1 - \mu_1^{-1} = 1/\varepsilon_1$:

$$\frac{\partial \log Y}{\partial \log \tau_1} = \frac{\sigma_1 \lambda_1 (\tilde{\lambda}_1 - 1)}{\varepsilon_1}, \quad \frac{\partial \log Y}{\partial \log M_1} = \eta_1 \lambda_1 + (\eta_1 + \xi_1) \frac{\sigma_1 \lambda_1 (\tilde{\lambda}_1 - 1)}{\varepsilon_1},$$

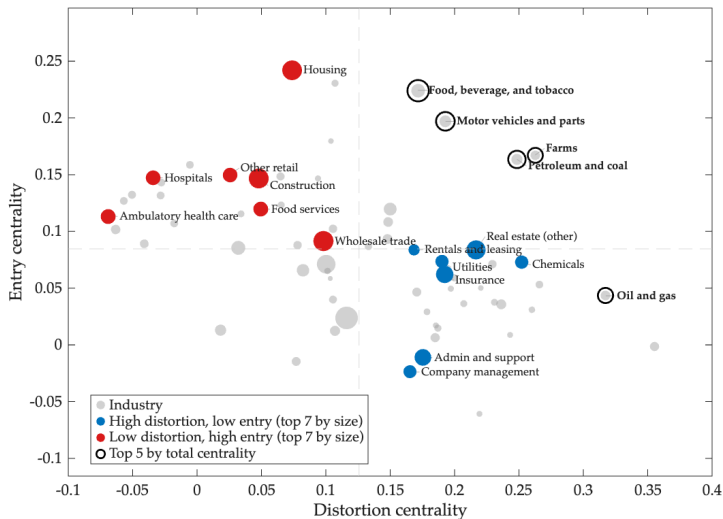
$$\frac{d \log M_1}{d \log \tau_1} = \frac{\sigma_1 (\lambda_1 - 1)}{1 + \eta_1 (1 - \sigma_1) (\lambda_1 - 1) + \xi_1 [(\varepsilon_1 - 1) - \sigma_1 (\lambda_1 - 1)]}$$

- Distortion centrality > 0 whenever the sector uses intermediates ($\tilde{\lambda}_1 > 1$) at a finite markup; scales with roundaboutness, independent of η_1, ξ_1
- Pro-competitive channel: raises the output value of each entrant, but dampens own entry ($(\varepsilon_1 - 1)\xi_1$: thinner margins fund fewer firms) unless the roundabout feedback $\sigma_1(\lambda_1 - 1)$ dominates
- Dixit–Stiglitz: entry response $\sigma_1(\lambda_1 - 1) > 0$ whenever the economy is roundabout ($\lambda_1 > 1$)

Calibration

- IO table: 2019 BEA summary input–output table, $N = 66$ sectors; labor as the single primary input
- Baseline within-sector aggregator: Dixit–Stiglitz with $\varepsilon_k = 8$ for all k
 - Pins down the desired markup $\varepsilon_k/(\varepsilon_k - 1)$, returns to variety $\eta_k = 1/(\varepsilon_k - 1)$, and $\xi_k = 0$ — a conservative benchmark that isolates the returns-to-variety channel
 - Uniform ε_k : divergence between centralities reflects *network position*, not imposed markup heterogeneity
- Cross-sector elasticities: $\sigma_0 = 0.9$ (consumption), $\sigma_k = 0.5$ (production) (Atalay, 2017)
- Robustness: $\xi_k \in [0, 0.3]$ disciplined by pass-through evidence, $\xi_k = (1/\rho_k - 1)(1 + \eta_k)$ (Baqae et al., 2024; Amiti et al., 2019)
- Both centralities reported per unit of sales (normalized by the Domar weight λ_k) — output response per unit of intervention cost, not a measure of size

Entry Centrality Re-Ranks U.S. Sectors



- $\text{corr}(\text{distortion}, \text{entry}) = -0.40$; upstreamness: $+0.89$ (distortion) vs. -0.76 (entry) — the two statistics load on opposite ends of the supply chain

The Two Margins Identify Different Sectors

- Highest entry centrality: housing, financial vehicles, food and beverage, motor vehicles — close to final demand (upstreamness 1–2), yet frequent policy targets
- Lowest entry centrality: forestry and fishing, company management, computers and electronics, primary metals — far upstream (upstreamness > 3), where distortion centrality is large
- Regressing total centrality on distortion centrality alone leaves 38% of the cross-sector variation unexplained $\Rightarrow$ the policy ranking cannot be recovered from the intensive margin by any monotone adjustment
- Downstream markets are central not because they supply many sectors, but because **expanding them mobilizes entry in the markets that supply them**

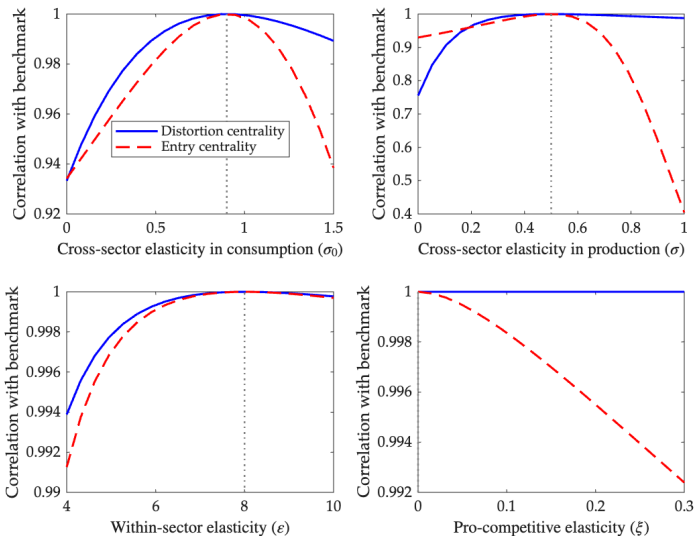
How Large Is the Extensive Margin?

- Levels of both centralities scale with the wedge; their *ratio* is pinned by the IO data. Share of the response to a uniform marginal subsidy operating through the extensive margin:

Within-sector elasticity	Pro-competitive elasticity ξ			
	0	0.1	0.2	0.3
$\varepsilon = 6$	0.44 (1.00)	0.37 (0.98)	0.33 (0.96)	0.30 (0.95)
$\varepsilon = 8$	0.45 (1.00)	0.36 (0.97)	0.30 (0.96)	0.27 (0.93)
$\varepsilon = 10$	0.45 (1.00)	0.34 (0.97)	0.28 (0.94)	0.24 (0.92)
$\varepsilon = 12$	0.46 (1.00)	0.32 (0.96)	0.25 (0.93)	0.22 (0.91)

- Between one quarter and one half** of the first-order output response operates through the margin an intensive-margin criterion omits (45% at the benchmark; 0.28 in China)
- Rank correlations (in parentheses) stay ≥ 0.91 : the re-ranking is unaffected

Robustness: Rankings Are Stable



- Correlation with the benchmark ranking as one parameter varies: both centralities load on σ_0, σ ; only entry centrality responds to ξ ; uniform changes in ϵ barely matter

Application: Sectoral Interventions in China

- Recalibrate with China's 2007 input–output table, 135 three-digit sectors, benchmark calibration
- Same pattern as the U.S.: $\text{corr}(\text{distortion}, \text{entry}) = -0.79$
 - High-distortion, low-entry: upstream energy and heavy materials (electric power, petroleum, coal, basic chemicals, steel)
 - High-entry, low-distortion: downstream services and primary activities (public administration, education, health care, real estate, trade)
- Do observed interventions load on the two statistics? Sectoral intervention measures from Liu (2019) (2007 Chinese Annual Survey of Industrial Firms, 79 manufacturing sectors): effective interest rates, debt-to-capital ratios, tax incentives of private firms

$$\text{Outcome}_i = \alpha + \beta \text{Distortion centrality}_i + \gamma \text{Entry centrality}_i + X_i' \delta + \epsilon_i$$

weighted by value added, with the Liu (2019) controls in X_i

- Descriptive, not causal: is the extensive-margin statistic redundant for observed policy?

Entry Centrality Helps Explain Observed Interventions

	Effective Interest Rate				Debt Ratio			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Distortion cent.	-2.04** (0.68)	-10.19*** (1.21)	-3.02*** (0.69)	-9.71*** (1.27)	7.77*** (1.70)	20.95*** (3.59)	8.00*** (1.94)	20.76*** (4.03)
Entry cent.		-16.46*** (2.20)		-14.58*** (2.48)		26.59*** (6.53)		27.81*** (7.85)
Liu (2019) controls	No	No	Yes	Yes	No	No	Yes	Yes
R-squared	0.104	0.485	0.343	0.556	0.214	0.355	0.262	0.371
Observations	79	79	79	79	79	79	79	79

- Both centralities predict more favorable financing; adding entry centrality raises R^2 substantially (interest rate: 0.343 $\rightarrow$ 0.556 with controls)

Conclusion

- Industrial policy changes **market structure**, not only wedges: sales, the mass of firms, varieties, and competitive pressure
- The output response to a sectoral subsidy = distortion centrality (intensive margin) + entry centrality (extensive margin: the value of the entry cascade, in closed form)
- The two statistics load on opposite ends of the supply chain and are negatively correlated in the U.S. and China; the extensive margin is between one quarter and one half of the aggregate response
- Evaluating sectoral policy requires combining *where distortions are located* with *how entry and competitive pressure propagate through the network*

References I

- Amiti, Mary, Oleg Itskhoki, and Jozef Konings (2019) 'International shocks, variable markups, and domestic prices.' *The Review of Economic Studies* 86(6), 2356–2402
- Atalay, Enghin (2017) 'How important are sectoral shocks?' *American Economic Journal: Macroeconomics* 9(4), 254–280
- Baqaei, David Rezza (2018) 'Cascading failures in production networks.' *Econometrica* 86(5), 1819–1838
- Baqaei, David Rezza, and Emmanuel Farhi (2020) 'Productivity and misallocation in general equilibrium.' *The Quarterly Journal of Economics* 135(1), 105–163
- Baqaei, David Rezza, Emmanuel Farhi, and Kunal Sangani (2024) 'The supply-side effects of monetary policy.' *Journal of Political Economy* 132(4), 1065–1112
- Bigio, Saki, and Jennifer La'O (2020) 'Distortions in production networks.' *The Quarterly Journal of Economics* 135(4), 2187–2253
- Edmond, Chris, Virgiliu Midrigan, and Daniel Yi Xu (2023) 'How costly are markups?' *Journal of Political Economy* 131(7), 1619–1675
- Elliott, Matthew, Benjamin Golub, and Matthew V Leduc (2022) 'Supply network formation and fragility.' *American Economic Review* 112(8), 2701–2747
- Liu, Ernest (2019) 'Industrial policies in production networks.' *The Quarterly Journal of Economics* 134(4), 1883–1948
- Taschereau-Dumouchel, Mathieu (2026) 'Cascades and fluctuations in an economy with an endogenous production network.' *Review of Economic Studies* 93(2), 1354–1392